

Real-Time Stock Viewer: Monitors live market prices for arbitrary stocks.

This demo uses the Datafeed Toolbox and either Bloomberg (requires a Bloomberg license) or Yahoo (requires an internet connection) to monitor market prices of various stocks. If the Datafeed Toolbox is not available, random numbers can be used instead.

The point of this demo is to show one way in which MATLAB can be used in a “real-time” environment. The only analysis being done is whether the prices have increased or decreased. If the price has decreased the point on the graph appears red. If it increases, it appears green. More analysis of the returned data could easily be added.

This demo contains seven files. Six are for command line operation, and one is a GUI that combines the previous six.

Two for Bloomberg – definebloomtimer.m, bloomRT.m

Two for Yahoo – defineyahootimer.m, yahooRT.m

Two for a randomly generated price – definerandomtimer.m, randomRT.m

To add your own analysis to the incoming data, replace or edit the <source>RT functions accordingly.

One GUI m-file – RTviewer.m lets you the data source, period, and ticker to update from a graphical user interface.

To add your own analysis functionality to the GUI, edit the <source>RT sub functions in the main RTviewer.m file.

Toolboxes: Datafeed

Command Line Instructions:

- 1) Depending on your data source, open the define<source>timer.m file to change the desired ticker to monitor and updating period. To run the code, enter the following command from the command prompt:

```
>> define<source>timer; start(t);
```

This will start the process, and update the stock price every period seconds, where period is defined in the define<source>timer.m file.

- 2) To stop the process, type the following at the command prompt:

```
>> stop(t),delete(t)
```

GUI Instructions:

- 1) Run the RTviewer.m file
- 2) Set the data source, ticker to monitor, and period to update.

Note: if using Bloomberg, you need to have a Bloomberg license and an internet connection. If using Yahoo, you need an internet connection. If using a random data source, the ticker only changes what is in the plot's title.

Make sure the ticker you enter is valid. Also, if using Bloomberg, only tickers in the equity market are supported. The market may be changed by editing the following line in the edtTicker_Callback radBloomberg_Callback sub functions

```
ticker = [ticker, ' US Equity'];
```

- 3) Press the “Run” toggle button to start and stop the process. The data will be plotted in the axes.